
WT9932P4 TINY User Guide

WT9932P4 Development Board User Manual

2025-08-6

目录

Disclaimer	2
Note	2
1. Development Board Overview	4
1.1 Product Classification	4
1.2 Development Board Introduction	4
1.3 Product Features	4
1.4 Development Board Image	5
2. Hardware Reference	6
2.1 Functional Block Diagram	6
2.2 Hardware Peripherals	7
2.3 Component Introduction	7
2.4 Interface Package	8
2.5 Interface Description	8
[J1] MIPI DSI FPC	8
[J2] MIPI CSI FPC	10
[J3] [J4] USB Power Interfaces	11
[J5] MicroSD Card Interface	11
[J6] [J7] Header Pins	12
3. Schematic Diagram	15
4. Structural Support	17
4.1 PCBA Dimension Diagram	17
5. Usage Guide	18
5.1 Preparation Steps	18
5.2 Hardware Setup	18
5.3 Usage Guide	18
6. Related Documentation	19
7. Contact Us	19

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Revision History

Version	Date	Changed By	Change Item
V1.0	2025/08/6	Kirto	Initial Document Creation



1. Development Board Overview

1.1 Product Classification

1.2 Development Board Introduction

The WT9932P4-TINY development board is centered around the Espressif ESP32P4 chip, making it a powerful tool for embedded development. Measuring only 69 × 28 mm, its compact design is well-suited for space-constrained applications.

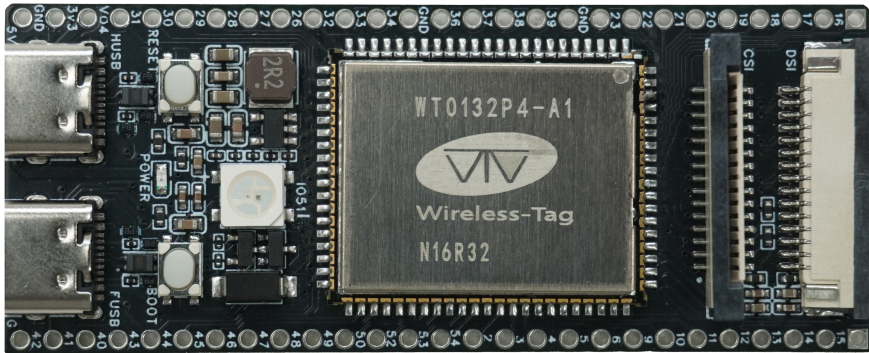
The board is based on the WT0132P4-A1 module and features a two-layer PCB design, ensuring stability while reducing cost. It comes with onboard MIPI DSI and CSI FPC interfaces, supporting display and image capture, making it ideal for applications such as security monitoring and robotic navigation.

All pins are fully exposed, and the header pins are compatible with breadboards, greatly enhancing expansion flexibility. The onboard RGB LED enables colorful status indications and interactive effects. Additionally, the board is equipped with Type-C FUSB and HUSB interfaces, offering fast data transfer and reversible connections, providing convenience for debugging and power supply. This makes it an ideal choice for IoT and other development scenarios.

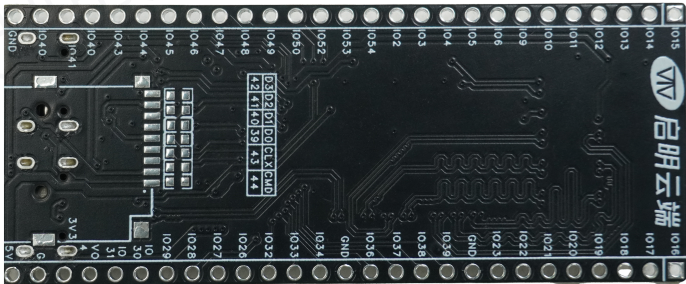
1.3 Product Features

1. **Core Configuration:** Powered by the high-performance Espressif ESP32P4 chip, meeting the performance requirements of embedded development.
2. **Compact Size:** Small form factor of only 69 × 28 mm, suitable for space-constrained applications.
3. **Stable Design:** Built on the WT0132P4-A1 module with a two-layer PCB design, ensuring stability while reducing cost.
4. **Image Interfaces:** Onboard MIPI DSI and CSI FPC interfaces support display output and image capture functions.
5. **Flexible Expansion:** All pins are exposed, and the header pins are compatible with breadboards, facilitating peripheral expansion.
6. **Status Indication:** Equipped with an onboard RGB LED for colorful status indications and interactive effects.
7. **Convenient Interfaces:** Features Type-C FUSB and HUSB ports, providing fast data transfer and reversible connections for easy debugging and power supply.

1.4 Development Board Image



Front View:



Rear View:

2. Hardware Reference

2.1 Functional Block Diagram

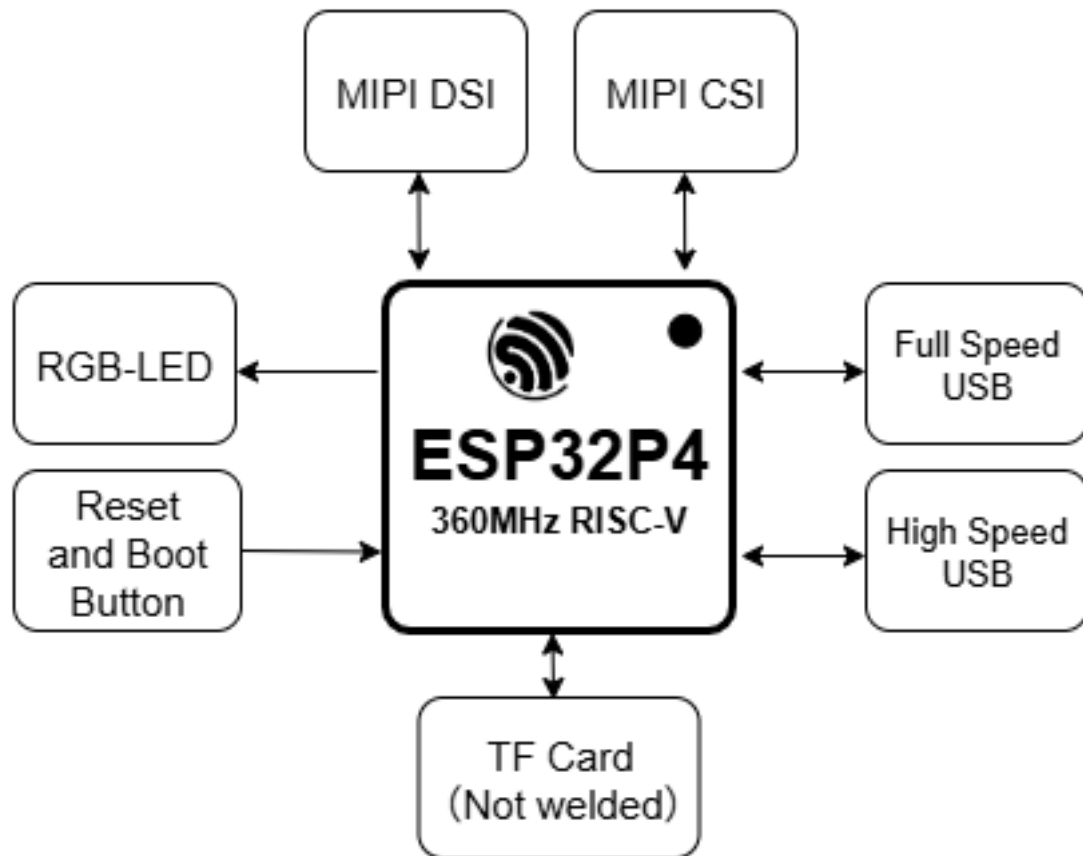


图 1 – Functional Block Diagram

2.2 Hardware Peripherals

Peripheral Name	Description
WS2812	Single-wire RGB LED connected to IO51.
RESET Button	Used for resetting the board.
BOOT Button	Used to manually enter flashing mode, connected to IO25.
FUSB	Full-speed USB, used for power and firmware updates.
HUSB	High-speed USB.
MIPI CSI FPC	1.0 pitch Raspberry Pi-compatible camera interface.
MIPI DSI FPC	1.0 pitch Raspberry Pi-compatible display interface.
MicroSD Card (not soldered)	4-wire MicroSD card slot, not soldered at factory.

2.3 Component Introduction

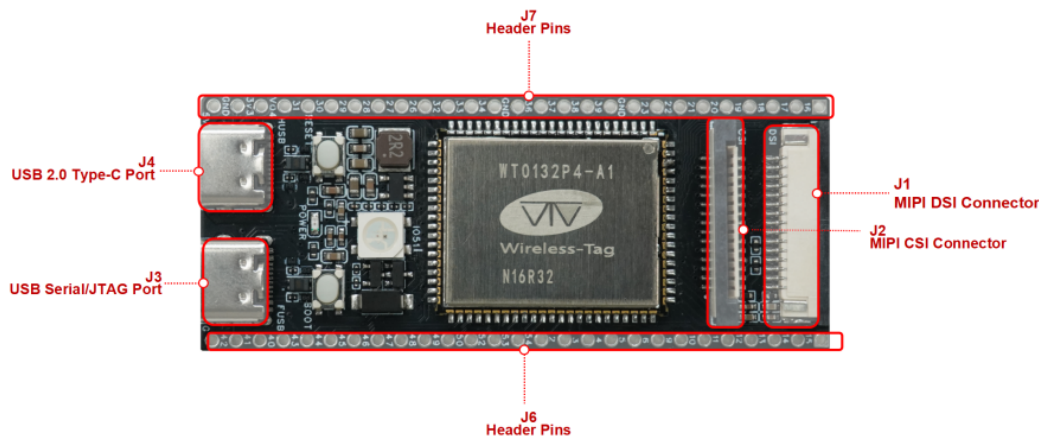


图 2 – Font Component Introduction

Component		
No.	Name	Description
J1	MIPI DSI Connector	FPC connector for connecting an LCD expansion board. Compatible FPC specifications: 1.0 mm pitch, 0.7 mm pin width, 0.3 mm thickness, 15 pins.
J2	MIPI CSI Connector	FPC connector for connecting an external camera module for image transmission. Compatible FPC specifications: 1.0 mm pitch, 0.7 mm pin width, 0.3 mm thickness, 15 pins.

Component		
No.	Name	Description
J3	USB Serial/ JTAG Port	USB Type-C interface supporting USB 2.0 Full-speed. Can be used to flash firmware to the ESP32-P4 chip, communicate via USB protocol, or perform JTAG debugging.
J4	USB 2.0 Type-C Port	USB 2.0 Type-A interface connected to the ESP32-P4 chip's USB 2.0 OTG High-Speed interface, supporting USB 2.0 standard. When used for USB communication, the ESP32-P4 acts as a USB Host and can supply up to 500 mA to connected USB devices.
J5	MicroSD Card Slot (Not Soldered)	Supports 4-bit mode MicroSD cards; not soldered at factory.
J6/ J7	Header Pins	All available GPIO pins are exposed to the headers J6 and J7. For detailed information, see Header Pins.

2.4 Interface Package

Interface Description	Interface Package	Note
MIPI DSI FPC	FPC_SMD_1.0K-FX-15PWBG	Use FPC with the same pin sequence
MIPI CSI FPC	FPC_SMD_15P-P1.00_HDGC_1.0K-GT-15PB	Use FPC with reverse pin sequence
MicroSD Card Interface	MicroSD-SMD_MicroSD-PUSH	Not soldered at factory
Header Pins x2	HDR-TH_27P-P2.54-V-M	Pitch supports perfboard/ breadboard

2.5 Interface Description

[J1] MIPI DSI FPC

The FPC has an SMD pitch of 1.0 mm with 15 pins, drawer-type, bottom-mounted, and lying flat. Please use an FPC with the same pin sequence. Diagram as shown:

FPC Pinout:

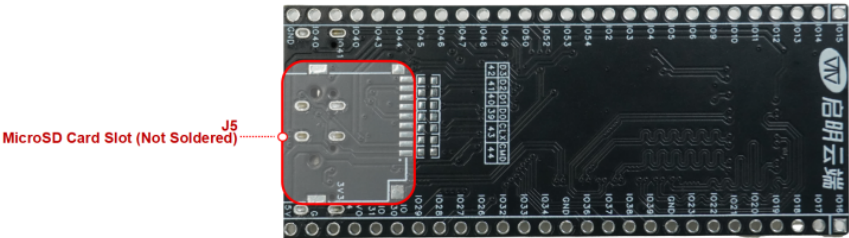


图 3 – Rear Component Introduction

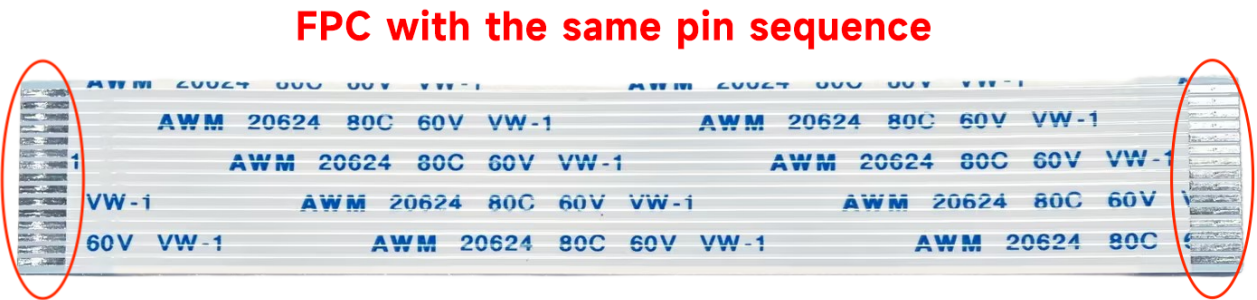


图 4 – Same Pin Sequence FPC

No.	Pin	Description	Voltage Range	Note
1	VCC	Power output	3.3V	Do not use as power input
2	IO7	I2C SDA pin	0V/3.3V	
3	IO8	I2C SCL pin	0V/3.3V	
4	IO0	Unused pin	0V/3.3V	
5	IO1	Camera reset pin	0V/3.3V	
6	GND	Ground	0V	
7	CSI_CLKP	CSI Clock positive differential	0V/3.3V	
8	CSI_CLKN	CSI Clock negative differential	0V/3.3V	
9	GND	Ground	0V	
10	CSI_DATAP1	CSI Data 1 positive differential	0V/3.3V	
11	CSI_DATAN1	CSI Data 1 negative differential	0V/3.3V	
12	GND	Ground	0V	
13	CSI_DATAP0	CSI Data 0 positive differential	0V/3.3V	
14	CSI_DATAN0	CSI Data 0 negative differential	0V/3.3V	
15	GND	Ground	0V	

[J3] [J4] USB Power Interfaces

USB 2.0 Type-C interfaces for firmware download and power supply.

- **J3:** Full-speed USB 2.0 with a maximum theoretical speed of **12 Mbps**, using IO24 and IO25.
- **J4:** High-speed USB 2.0 with a maximum theoretical speed of **480 Mbps**, using dedicated IO pins.

[J5] MicroSD Card Interface

Spring-loaded MicroSD card slot:

ⓘ Note: The power here uses LDO_VO4 and can be configured to support 1.8V supply.

No.	Pin	Description	Voltage Range	Note
1	IO41	DATA2 in 4-wire SDIO	0V / LDO_VO4	
2	IO42	DATA2 in 4-wire SDIO	0V / LDO_VO4	
3	IO44	DATA2 in 4-wire SDIO	0V / LDO_VO4	
4	LDO_VO4	SD card power supply	LDO_VO4	
5	IO43	DATA2 in 4-wire SDIO	0V / LDO_VO4	
6	GND	Ground	0V	
7	IO39	DATA2 in 4-wire SDIO	0V / LDO_VO4	
8	IO40	DATA2 in 4-wire SDIO	0V / LDO_VO4	

[J6] [J7] Header Pins

The header pins are standard 2.54 mm pitch.

[J6] Pin Definition:

No.	Pin	Description	Voltage Range	Note
1	IO15	General-purpose I/O 15	0V/3.3V	
2	IO14	General-purpose I/O 14	0V/3.3V	
3	IO13	General-purpose I/O 13	0V/3.3V	
4	IO12	General-purpose I/O 12	0V/3.3V	
5	IO11	General-purpose I/O 11	0V/3.3V	
6	IO10	General-purpose I/O 10	0V/3.3V	
7	IO9	General-purpose I/O 9	0V/3.3V	
8	IO6	General-purpose I/O 6	0V/3.3V	
9	IO5	General-purpose I/O 5	0V/3.3V	Also JTAG MTCK
10	IO4	General-purpose I/O 4	0V/3.3V	Also JTAG MTDI
11	IO3	General-purpose I/O 3	0V/3.3V	Also JTAG MTMS
12	IO2	General-purpose I/O 2	0V/3.3V	Also JTAG MTCK
13	IO54	General-purpose I/O 54	0V/3.3V	
14	IO53	General-purpose I/O 53	0V/3.3V	

No.	Pin	Description	Voltage Range	Note
15	IO52	General-purpose I/O 52	0V/3.3V	
16	IO50	General-purpose I/O 50	0V/3.3V	
17	IO49	General-purpose I/O 49	0V/3.3V	
18	IO48	SD Card DATA7	0V/LDO_VO4	Can also be used as GPIO
19	IO47	SD Card DATA6	0V/LDO_VO4	Can also be used as GPIO
20	IO46	SD Card DATA5	0V/LDO_VO4	Can also be used as GPIO
21	IO45	SD Card DATA4	0V/LDO_VO4	Can also be used as GPIO
22	IO44	SD Card CMD	0V/LDO_VO4	Can also be used as GPIO
23	IO43	SD Card CLK	0V/LDO_VO4	Output only if used as GPIO
24	IO40	SD Card DATA1	0V/LDO_VO4	Can also be used as GPIO
25	IO41	General-purpose I/O 41	0V/3.3V	
26	IO42	General-purpose I/O 42	0V/3.3V	
27	GND	Ground	0V	

[J7] Pin Definition:

No.	Pin	Description	Voltage Range	Note
1	IO16	General-purpose I/O 16	0V/3.3V	
2	IO17	General-purpose I/O 17	0V/3.3V	
3	IO18	General-purpose I/O 18	0V/3.3V	
4	IO19	General-purpose I/O 19	0V/3.3V	
5	IO20	General-purpose I/O 20	0V/3.3V	
6	IO21	General-purpose I/O 21	0V/3.3V	
7	IO22	General-purpose I/O 22	0V/3.3V	
8	IO23	General-purpose I/O 23	0V/3.3V	
9	GND	Ground	0V	
10	IO39	SD Card DATA0	0V/ LDO_VO4	Can also be used as GPIO

No.	Pin	Description	Voltage Range	Note
11	IO38	UART0 RXD for debugging or flashing	0V/3.3V	Strapping pin
12	IO37	UART0 TXD for debugging or flashing	0V/3.3V	Strapping pin
13	IO36	General-purpose I/O 36	0V/3.3V	Strapping pin
14	GND	Ground	0V	
15	IO34	General-purpose I/O 34	0V/3.3V	Strapping pin
16	IO33	General-purpose I/O 33	0V/3.3V	
17	IO32	General-purpose I/O 32	0V/3.3V	
18	IO26	General-purpose I/O 26	0V/3.3V	
19	IO27	General-purpose I/O 27	0V/3.3V	
20	IO28	General-purpose I/O 28	0V/3.3V	
21	IO29	General-purpose I/O 29	0V/3.3V	
22	IO30	General-purpose I/O 30	0V/3.3V	
23	IO31	General-purpose I/O 31	0V/3.3V	
24	LDO_VO4	SD Card Power	LDO_VO4	
25	3.3V	3.3V Power Output	3.3V	Do not input 3.3V
26	GND	Ground	0V	
27	5V	5V Power	5V	If both J3 and J4 are connected, J3 takes priority for power

☒ Note: After reset, the strapping pin values determine whether the board enters flashing mode.

Boot Mode	GPIO35	GPIO36	GPIO37	GPIO38
SPI Boot	1	Any	Any	Any
Joint Download Boot	0	1	Any	Any

3. Schematic Diagram

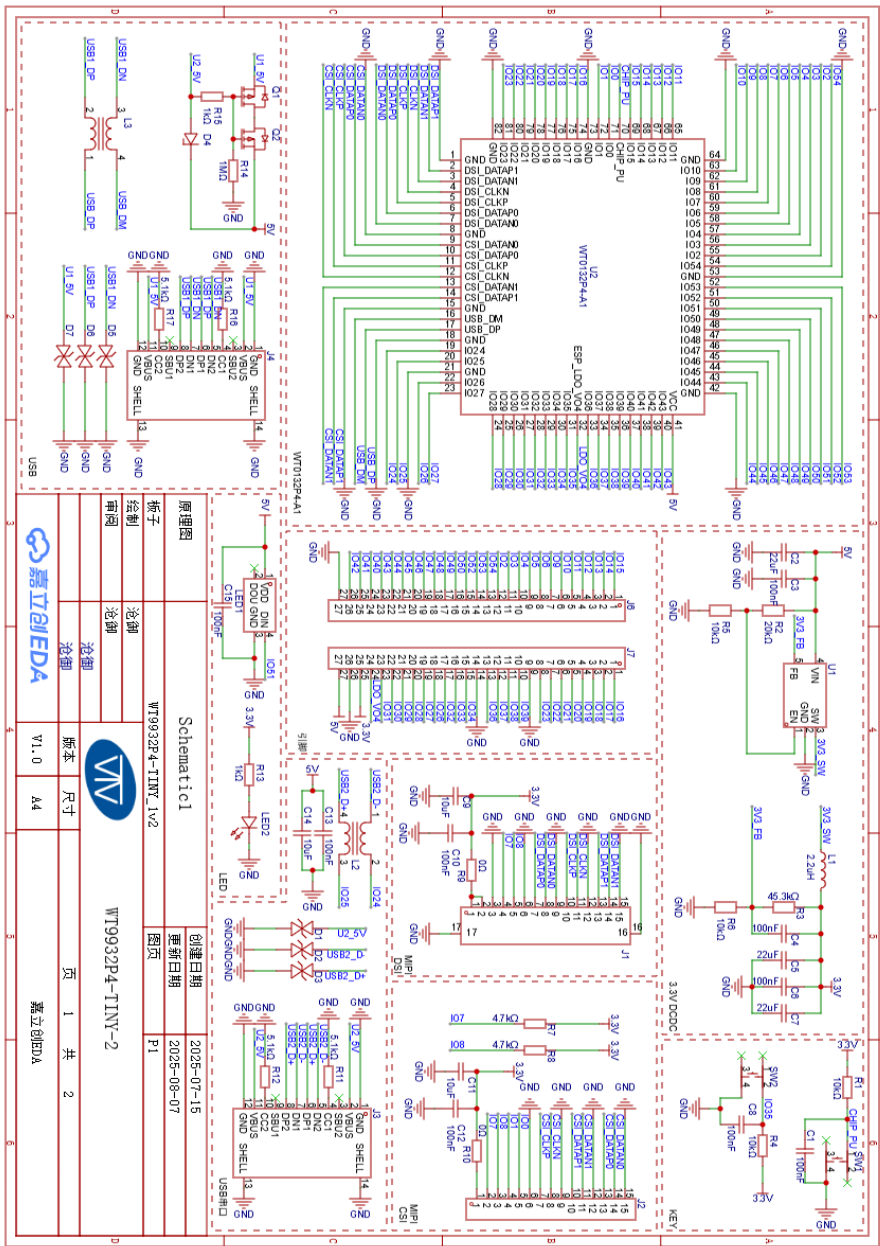


图6 – Schematic Diagram



4. Structural Support

4.1 PCBA Dimension Diagram

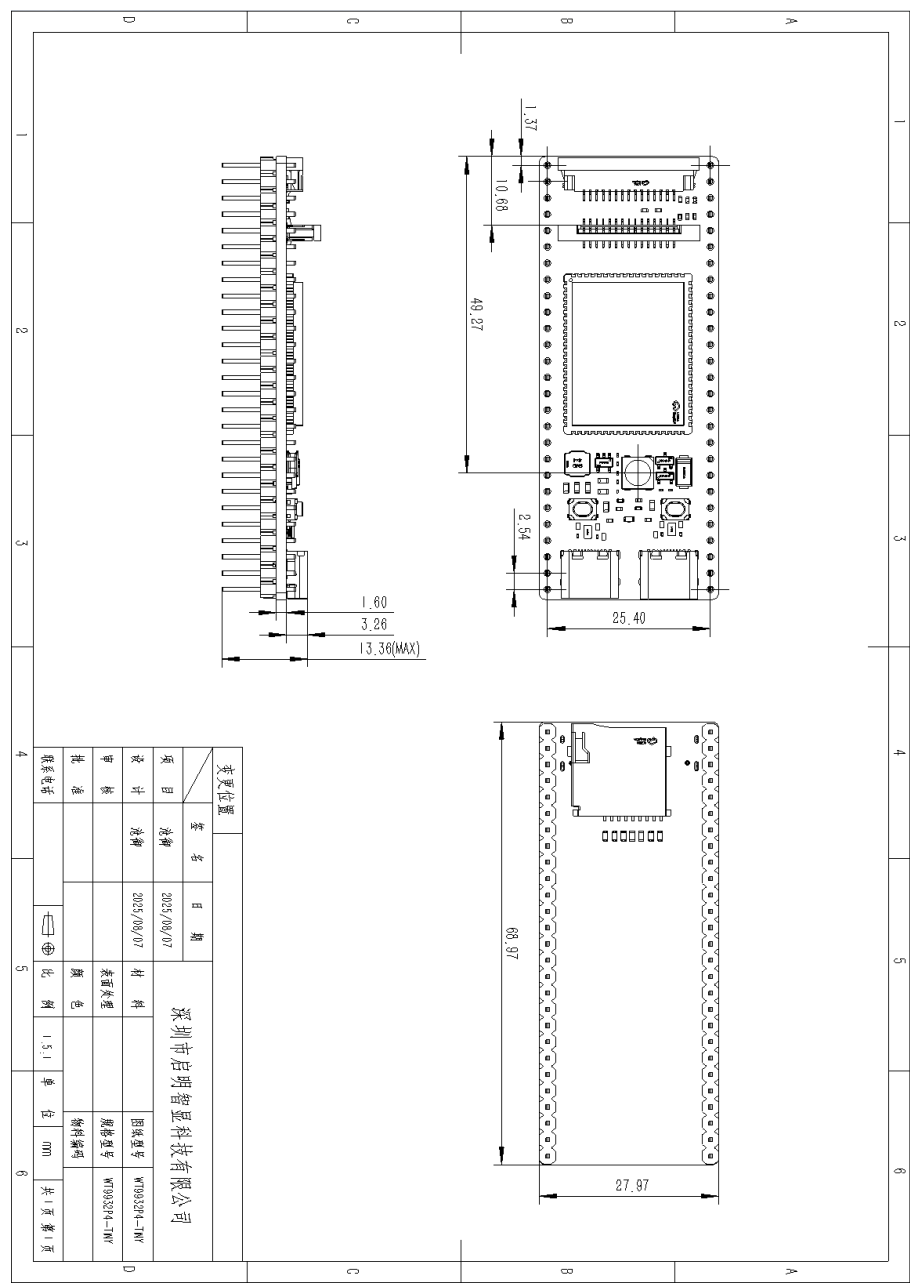


图 8 – Main Board Dimension Diagram

5. Usage Guide

5.1 Preparation Steps

Before using the device, please prepare the following items and environmental conditions:

1. **USB-C Data Cable** × 1 *Used for powering the development board or connecting to a computer. It is recommended to use the original or a high-quality cable to ensure stable power supply.*
2. **WT9932P4-TINY Development Board** × 1 *The core hardware device of this product.*

Tip: If using USB power from a computer, ensure the USB port can provide at least 5V@1A; insufficient power may prevent the device from starting properly.

5.2 Hardware Setup

1. Connect Power

- Use the USB-C data cable to connect the development board to a computer or USB adapter.
- Recommended adapter specification: **5V/1A or higher** to ensure stable power.
- Once powered on, the device will start automatically, and the LED indicator will light up to show the board is powered.

5.3 Usage Guide

1. **Set up the official ESP-IDF environment** [Official Quick Start](#)
2. **Clone Example Projects** Clone the official blink example and export the ESP-IDF environment.
3. **Build and Flash the Example** Connect the development board via USB-C and execute:

```
1 idf.py set-target esp32p4
2 idf.py build
```

To compile the example, then flash it using:

```
1 idf.py flash
```

After flashing, the RGB LED on the development board should blink, indicating successful operation.

6. Related Documentation

Document		
Name	Link	Description
WT9932P4-TINY User Guide	WT9932P4-TINY User Guide	
ESP32-P4 IDF Online Documentation	https://docs.espressif.com/projects/esp-idf/zh_CN/v5.4.2/esp32p4/get-started/index.html	ESP-IDF programming guide and API reference
ESP32-P4 Technical Reference Manual	https://www.espressif.com.cn/sites/default/files/documentation/esp32-p4_technical_reference_manual_cn.pdf	ESP32-P4 chip specification reference

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